Stop interpolation Topologically

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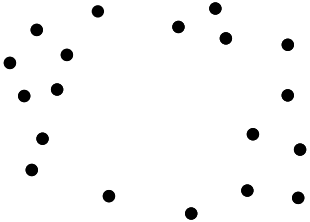
Topology matters

Guess the manifold



This set of points forms some shape. It can be considered as an **annulus**.
The data set shall be augmented by additional points preserving the shape.
How can we guarantee this?

Simplices

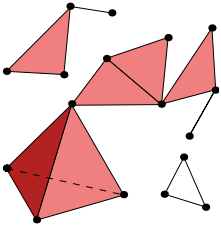
Given a set $X = \{x_0, \cdots, x_k\} \subset \mathbb{R}^d$ of $k+1$ points that do not lie on a hyperplane with dimension less than d, the k-dimensional simplex σ spanned by X is the set of convex combinations, such that

$$\sum_{i=0}^{k} \lambda_i x_i \quad \text{with} \quad \sum_{i=0}^{k} \lambda_i = 1 \quad \text{and} \quad \lambda_i \geq 0.$$

Simplicial complexes

A **simplicial complex** K is a collection of simplices such that:

- Any face of a simplex of K is a simplex of K.
- Any intersection of two simplices in K is either empty or a common face of both.



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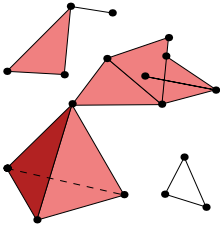

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Persistent Homology

An outline to persistent homology

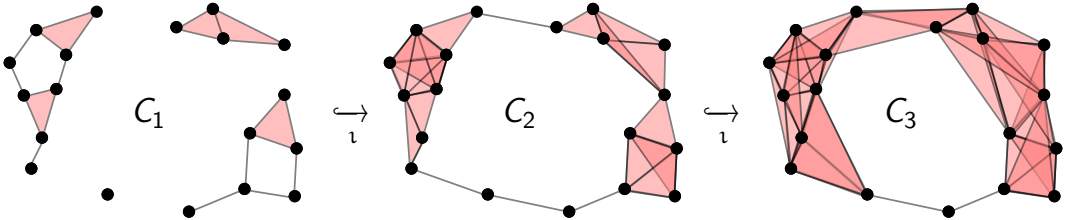
A brief description

Persistent homology is the study of k-**dimensional abelian groups** attached to a topological space. They are studied algebraically and behave *similarly* to the topological space itself under change revealing properties of the original space they are attached to by their *invariants*.

The connection of abelian groups by a boundary homomorphism allows one to investigate all k-dimensions at once. The induced object is a chain complex.

Chain complex

An example



Chain complex

How does it work?

Homology theory is the study of properties of an object X, such as a set of points, by means of homological (commutative) algebra. One assigns to X a sequence of abelian groups/modules $C_0, C_1, \cdots$ which are connected by homomorphisms $\partial_{k+1} : C_{k+1} \longrightarrow C_k$ such that $\mathrm{im}\partial_{k+1} \subseteq \ker \partial_k$. Such a structure is called **chain complex**.

$$0 \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k-1}} C_{k-2} \xrightarrow{\partial_{k-2}} \cdots \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} 0$$

Homology groups

The kth **homology group** of a simplicial complex is defined as

$$H_k(K) = \frac{\ker \partial_k C_k(K)}{\text{im } \partial_{k+1} C_{k+1}(X)}.$$

Intuitively, the kernel of the boundary homomorphism of the kth chain group gives all k-cycles, thus the cycle group, from which we quotient out all elements of the kth boundary group, i.e.

$$H_k(K) = Z_k(K)/B_k(K).$$

Betti numbers

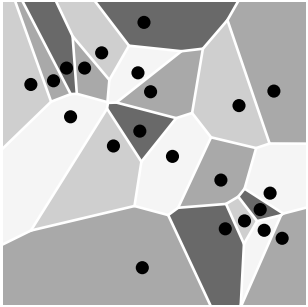
The kth **Betti** number β_k is the rank of the kth homology group $H_k(K)$ of the topological space K.

We'll track the betti numbers to track the amount of holes along the filtration.

Natural neighbor interpolation

Voronoi diagram

Natural neighbor interpolation example

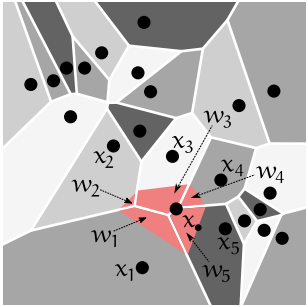


The **Voronoi Diagram** is spanned over the point cloud embedded into some Euclidean space. Technically one does not calculate the Voronoi diagram but the dual **Delaunay triangulation**.

$$\text{Vordgm} = \{x \in X \mid d(x, \lambda_i) \leq d(x, \lambda_j)\}.$$

Coordinate shift of the added point

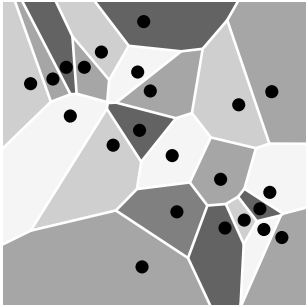
Weighting by Voronoi regions

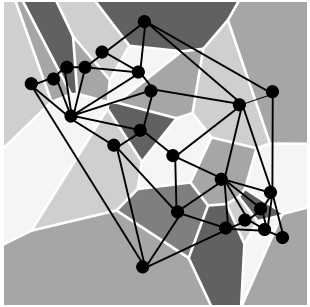


The algorithm re-weights the **coordinates of a new point** within the convex hull of a set of points by the change of Voronoi regions relative to the Voronoi regions without the added point:

$$\hat{x}^\bullet = \sum_{i=1}^{n} w_i x_i^\bullet. \tag{1}$$

Duality of Delaunay complex and Voronoi diagram





The Delaunay complex is spanned, connecting each point to its neighbors.
It is always well defined and does not create simplicial complexes which
can only be embedded into higher dimensions than the point set.

The collapsing sequence
Theorem of simplicial collapse

Theorem

Let the simplicial collapse $\searrow$ denote the removal of all σ''-simplices, where $\sigma \subseteq \sigma'' \subseteq \sigma'$, with σ being a free face. If X is a finite set of points in general position in some metric space, then

$$\check{\mathrm{C}}\mathrm{ech}(X, r) \searrow \mathrm{Del}\check{\mathrm{C}}\mathrm{ech}(X, r) \searrow \mathrm{Del}(X, r) \searrow \mathrm{Wrap}(X, r).$$

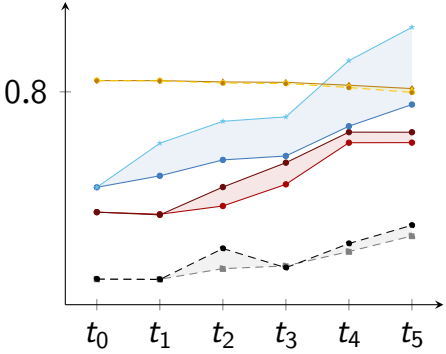
Established by Bauer and Edelsbrunner: The Morse theory of Čech and Delaunay complexes.

Results

Dataset description

- Investigation of 83 handwritings from different people.
- 45 signatures are recorded per user, which show the same letters.
- Each iteration we double the amount of data points.
- We insert the new points random uniformly within the convex hull.

Summary statistics



With a hypothesis test, calculated using the Wasserstein metric, we have defined a stop criterion. To validate the stop criterion, we calculated elmentary statistics on the whole dataset during interpolation.

Upper curve: no stopping.
Lower curve: with stopping.

- ●: Variation.
- ●: Standard deviation.
- ●: Mean.
- ●: 1-Wasserstein metric.

Summary

What do we have achieved?

- Connection of Collapsing Theorem to Voronoi interpolation.
- A mathematical stopping condition for interpolation techniques.
- A hypothesis test for persistence diagrams.

Open problems?

- What if points do not lie in general position with respect to $\mathbb{R}^n$?
- Which embedding should be used for Voronoi interpolation?
- Does Voronoi interpolation preserve homology groups?

Thank you for your attention!

Got interested?

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